

Insurance Customer Analytics Project (SQL)

Project Overview

This project analyzes an **insurance customer dataset** to extract meaningful business insights using **SQL (MySQL)**. The goal is to understand customer behavior, profitability, claims patterns, and revenue contribution across different segments such as state, policy type, coverage, and demographics.

The project simulates a real-world **insurance analytics use case** used by data analysts in business decision-making.

Dataset Information

Table Name: customer_insurance

Key Columns:

- Customer, State, Gender, Education, Marital_Status
 - Income, EmploymentStatus
 - Policy_Type, Policy, Coverage
 - Monthly_Premium_Auto
 - Total_Claim_Amount
 - Customer_Lifetime_Value
 - Number_of_Policies
 - Months_Since_Last_Claim
 - Number_of_Open_Complaints
 - Vehicle_Class, Vehicle_Size
 - Sales_Channel
 - Effective_To_Date_New
-

Data Preparation

- Renamed raw table to customer_insurance
 - Converted date column using STR_TO_DATE
 - Dropped old date column after transformation
 - Ensured structured and clean dataset for analysis
-

Exploratory Data Analysis (EDA)

Customer Insights

- Total number of customers
- Customers by state, gender, education
- Average income of customers
- Distribution of vehicle classes

Financial Metrics

- Average Customer Lifetime Value (CLV)
 - Highest and lowest monthly premium
 - Total policies sold
-



Business Analysis



State & Revenue Analysis

- State with highest premium revenue
- Total claim amount by state
- Average premium by state



Sales Channel Performance

- Sales channel generating highest revenue



Vehicle Analysis

- Vehicle class with highest claim amount
- Claim patterns by vehicle size



Policy Insights

- Policy type with highest CLV
 - Number of policies sold by customers
 - Most profitable policy types
-



Advanced SQL Analysis



Customer Segmentation

- Income-based segmentation (Low / Middle / High income)
- Income quartiles using NTILE()



Profitability Analysis

- Customer profitability = CLV – Total Claims
- Ranked customers from most to least profitable
- Identified top high-value customers

Risk & Behavior Analysis

- Customers likely to churn based on:
 - Complaints
 - Policy age
 - Response behavior
- Customers with higher risk profiles

Window Function Analysis

- Ranking customers within states by CLV
 - Running total of claims
 - Percentage contribution to premium revenue
-

Key Business Insights

- Certain states contribute significantly higher premium revenue.
 - A small group of customers generates the majority of profitability.
 - Customers with high complaints and low engagement show churn risk.
 - Vehicle class and policy type strongly influence claim behavior.
 - High-income customers contribute disproportionately to revenue.
-

Key SQL Concepts Used

- SELECT, WHERE, GROUP BY, ORDER BY
- Aggregate Functions (SUM, AVG, COUNT)
- Subqueries
- CTE (Common Table Expressions)
- Window Functions:
 - RANK()
 - ROW_NUMBER()
 - NTILE()
 - SUM OVER()

❏ Conclusion

This project demonstrates how SQL can be used to transform raw insurance data into **business-ready insights**. It covers customer segmentation, revenue analysis, risk identification, and profitability evaluation—key areas for data-driven decision-making in the insurance industry.

🚀 Tools Used

- MySQL
- SQL (Advanced + Window Functions)

🖼️ Must-have Screenshots

The screenshot shows a SQL IDE with two queries and their results. The first query calculates the annual premium revenue by state, and the second query calculates the total customer lifetime value by policy type.

```
-- Business Problems
-- Which state generates the highest premium revenue?
SELECT
    state,
    SUM(Monthly_Premium_Auto * 12) AS annual_premium_revenue
FROM customer_insurance
GROUP BY state
ORDER BY annual_premium_revenue DESC;

-- Which policy type is the most profitable based on customer lifetime value?
SELECT
    Policy_Type,
    SUM(Customer_Lifetime_Value * 12) AS total_clv
FROM customer_insurance
GROUP BY Policy_Type
ORDER BY total_clv DESC;
```

state	annual_premium_revenue
California	3534120
Oregon	2926560
Arizona	1877760
Nevada	997848
Washington	881292

The screenshot shows a SQL IDE with a complex query and its results. The query creates a database, uses it, renames a table, alters it to add a column, updates it, and then selects data from it.

```
1 create database InsuranceDB;
2 use InsuranceDB;
3
4 rename table 'ins_fn-uscc-marketing-customer-value-analysis' to 'customer_insurance';
5
6 ALTER TABLE customer_insurance
7 ADD COLUMN Effective_To_Date_New DATE;
8
9 UPDATE customer_insurance
10 SET Effective_To_Date_New = STR_TO_DATE('Effective To Date', '%d-%m-%Y');
11
12 SELECT 'Effective To Date'
13 FROM customer_insurance
14 LIMIT 10;
15
16 alter table customer_insurance drop column 'Effective To Date';
17
18 -- Basic SQL
```

Customer	State	Customer_Lifetime_Value	Response	Coverage	Education	EmploymentStatus	Gender	Income	Location Code	Marital_Status	Monthly_Premium_Auto	Months_Since_Last_Claim	Months_Since_Pol
BU79786	Washington	2763.519279	No	Basic	Bachelor	Employed	F	56274	Suburban	Married	69	32	5
QI40356	Arizona	6979.535983	No	Extended	Bachelor	Unemployed	F	0	Suburban	Single	94	13	42
A149188	Nevada	12887.43165	No	Premium	Bachelor	Employed	F	48767	Suburban	Married	108	18	38
WW63253	California	7645.861827	No	Basic	Bachelor	Unemployed	M	0	Suburban	Married	106	18	65
H64268	Washington	2813.692575	No	Basic	Bachelor	Employed	M	43836	Rural	Single	73	12	44
QC33172	Oregon	8236.3778	Yes	Basic	Bachelor	Employed	F	62992	Rural	Married	69	14	94
178711A	Oregon	1780.808636	Yes	Rate	Follow	Freelance	F	11100	Suburban	Married	67	0	17

queries* x customer_insurance

Save the script to a file.

```
273 )
274 AND COUNT(Policy) = 1;
275
276 -- Identify customers who may be at risk of churning based on policy age, cla
277 • SELECT
278     Customer,
279     State,
280     Policy_Type,
281     Months_Since_Last_Claim,
282     Number_of_Open_Complaints,
283     Number_of_Policies,
284     Response
285 FROM customer_insurance
286 WHERE
287     Number_of_Open_Complaints > 0
288     AND Response = 'No'
289     AND Months_Since_Last_Claim > 6
290     AND Number_of_Policies = 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

queries* x customer_insurance

```
273 )
274 AND COUNT(Policy) = 1;
275
276 -- Identify customers who may be at risk of churning based on policy age, claims, and response.
277 • SELECT
278     Customer,
279     State,
280     Policy_Type,
281     Months_Since_Last_Claim,
282     Number_of_Open_Complaints,
283     Number_of_Policies,
284     Response
285 FROM customer_insurance
286 WHERE
287     Number_of_Open_Complaints > 0
288     AND Response = 'No'
289     AND Months_Since_Last_Claim > 6
290     AND Number_of_Policies = 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Customer	State	Policy_Type	Months_Since_Last_Claim	Number_of_Open_Complaints	Number_of_Policies	Response
▶	EJ77678	Oregon	Personal Auto	17	1	1	No
	QG25316	Nevada	Corporate Auto	7	1	1	No
	ML29312	Oregon	Personal Auto	34	2	1	No
	OO88645	California	Corporate Auto	10	2	1	No
	II62831	California	Special Auto	12	1	1	No
	TR85083	California	Personal Auto	27	1	1	No
	DE55857	Washington	Personal Auto	7	1	1	No
	LA97014	Washington	Personal Auto	19	1	1	No
	LA97014	Washington	Special Auto	25	2	1	No

customer_insurance 20 x

queriescustomer_insurance

Don't Limit

```

296 • select EmploymentStatus , round(avg(total_claim_amount),2) as highest_average_claim_amount from customer_insurance group by EmploymentStatus order by highest_average_cl
297
298 -- Create a customer profitability report by comparing customer lifetime value with total claim amount and rank customers from most profitable to least profitable.
299 • SELECT
300     Customer,
301     Customer_Lifetime_Value,
302     Total_Claim_Amount,
303     (Customer_Lifetime_Value - Total_Claim_Amount) AS profitability,
304
305
306     RANK() OVER (
307         ORDER BY (Customer_Lifetime_Value - Total_Claim_Amount) DESC
308     ) AS profit_rank
309
310 FROM customer_insurance;
311
312
313 -- or
  
```

Result Grid

Filter Rows:

Exports

Wrap Cell Contents

Fetch rows:

	Customer	Customer_Lifetime_Value	Total_Claim_Amount	profitability	profit_rank
▶	FQ61281	83325.38119	1108.8	82216.58119	1
	YC54142	74228.51604	1742.4	72486.11604000001	2
	BP23267	73225.95652	969.6	72256.35652	3
	KH55886	67907.2705	151.711475	67755.559025	4
	SK66747	66025.75407	1194.892002	64830.862067999995	5
	FB95288	64618.75715	1562.4	63056.357149999996	6
	AZ84403	61850.18803	1142.4	60707.788029999996	7
	JT47995	60556.19213	979.2	59576.992130000006	8

Result 21 x

Output

Author

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